University of Central Punjab

Faculty of Information Technology and Computer Science

Course Title: Digital Logic and Design
Course Code: AR102

Assignment 1

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Section: B-17

CLO #	Course Learning Outcome (CLO)	Taxonomy Level	Mapping to PLO
CLO 1	Acquire knowledge related to the concepts, tools, and techniques to demonstrate the response of digital electronic circuits.	C3	1

Instructions:

1. Attempt all questions.
2. Write your answer showing all steps required to perform the task.
3. Assignment must be solved on A4 sheets or Assignment sheets only. Violation will result to deduction of 1 mark from the scored marks. Submissions will be made either via portal or in the class.
4. Each student will have attached this front page with his/her assignment. Violation will result to deduction of 1 mark from the scored marks.
5. Due Date for Assignment is: Next Class
6. Late submission will result in 10% deduction in marks.
7. No request for late submissions will be considered after two working days of the deadline.
8. No request to review assignment will be considered after 3 working days of the review in class.

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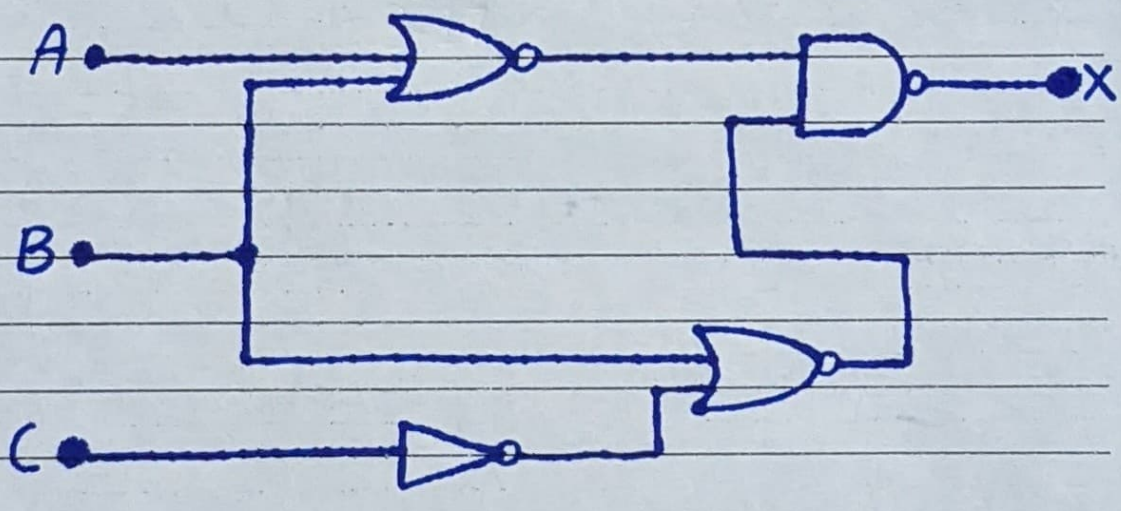
NAME: Zarar ROLL No: 0109 SUBJECT: DLD Date: 5/4/25

★ Problem # 1

Inter the output timing waveform of the circuit given below;

Circuit Diagram



★ Requirement for generating output timming waveform:-

We need waveform of three inputs A, B and C for generating output timming waveform X.

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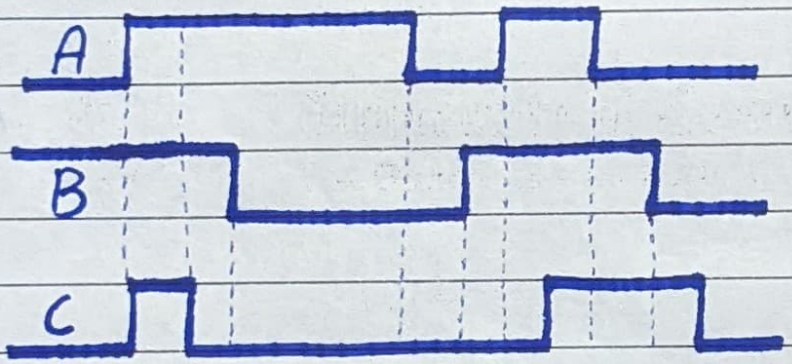
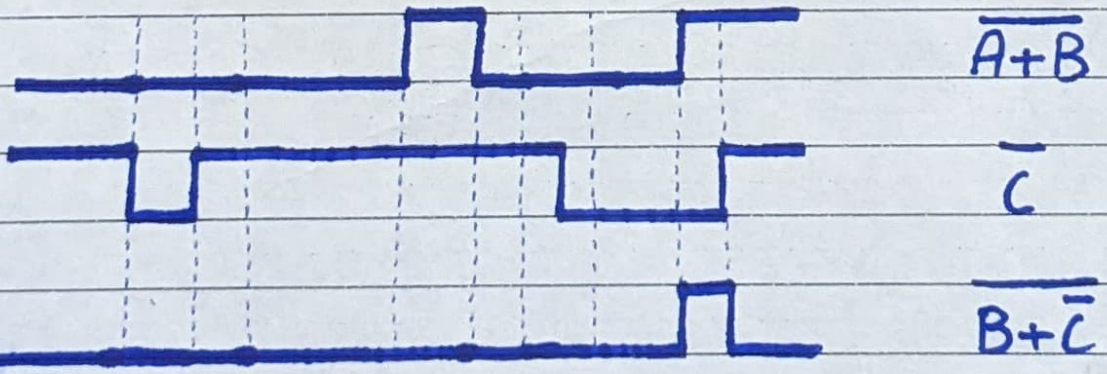
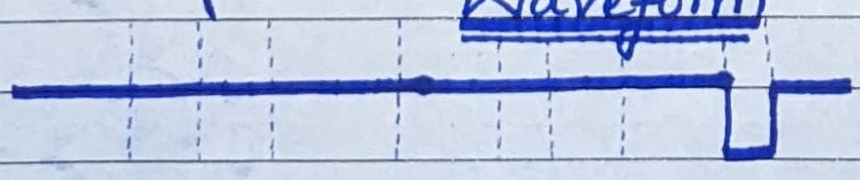
(Timming waveform as input)



(Timming waveform as output)



(Final output timming waveform)



$\boxed{(\overline{A+B})(\overline{B+\overline{C}})}$

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★ **Explanation of procedure**

For obtaining "output timming waveform", we need two individual waveform. First one is waveform of $\overline{A+B}$ (By applying NOR gate on input A and input B). Second one is waveform of $\overline{B+\overline{C}}$ (By applying NOR gate on input B and input $\overline{C}$)

NOR gate functionality

In OR gate, at least one input must be "1" for generating output "1". NOR gate reversed the functionality of OR gate by applying Not gate.

After obtaining waveform of $\overline{A+B}$ $\overline{B+\overline{C}}$, apply NAND gate.

NAND gate functionality

In AND gate, output is "1", when all inputs are "1".

★ **Final result**

$$\boxed{(\overline{A+B})(\overline{B+\overline{C}}) = X}$$

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Assignment: 1st Sheet No: 4 Date: 5 April 2025

* Problem # 2):-

Solve the hexadecimal addition and subtraction of the following numbers.

:(Adding):-

```
   ①
   2 A C 6
+  8 4 D 6
  -------
   A F 9 C
```

→ Step 1:-

6+6 = 12 < 16 (hexadecimal base), so we write value "12" variable.

$\boxed{C = 12}$

→ Step 2:-

value of C is "12" and value of D is "13". Then, C + D = 12 + 13, = 25

25 > 16 (hexadecimal base), so we use

(L.C.M method)

16	25
	1 — 9

write right side value (in this case 9) as answer and move left side value (in this case 1) as carry further.

→ Step 3:-

carry value $\boxed{1}$ + Variable 'A' value $\boxed{10}$ + $\boxed{4}$
= 15 < 16 (hexadecimal base), so we write variable; whose value is 15

$\boxed{F = 15}$

→ Step 4:-

8 + 2 = 10 < 16 (hexadecimal base), so we write variable; whose value is 10.

Assignment: 1st Sheet No: 6 Date: 5 April 2025

$\boxed{A = 10}$

★ Final result:-

After doing / following these four steps, final result is;-

$\boxed{AF9C}$

Subtraction

$\boxed{7}$	$\boxed{16+2}$	$\boxed{16+9}$	$\boxed{16+1=17}$
8	3	A	1
− 1	D	D	6
6	5	C	B

→ Step 1:-

We can't subtract $\boxed{6}$ from $\boxed{1}$, so we take carry from neighbor digit, which is variable A, whose value equals to $\boxed{10}$.

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Assignment: 1st Sheet No: 7 Date: 5 April 2025

Subtraction related to hexadecimal number system, so carry we take from neighbor digit is 16 (hexadecimal base). Which is actually equal to 1.

Now

(carry) $\boxed{16}+1=17$;

$17-6=11$

"11" is equals to variable B. So, **B** is written as <u>answer</u>

→ Step 2:-

We already take '1' as carry from variable A for previous value subtraction. Now;

remaining value is:- 9

We can't subtract variable 'D' value 13 from 9, so we take carry from neighbor digit; which is 3.

Subtraction related to hexadecimal number system, so carry we take from neighbor digit is 16 (hexadecimal

number system), which is actually equals to 1.

Now;-

(carry value) $\boxed{16}$ + (remaining value) $\boxed{9}$ = 25
{D = 13}

∴ 25 - 13 = 12

"12" equals to variable c. So, C is written as <u>answer</u>.

→ Step 3:-

Similarly, like previous two steps:-

(carry value) $\boxed{16}$ + remaining value $\boxed{2}$ = 18
{D = 13}

∴ 18 - 13 = 5

5 is written as <u>answer</u>.

→ Step 4:-

(remaining value) $\boxed{7}$ - 1 = 6

6 is written as <u>answer</u>.

* final answer

After following these four steps, resultant answer is;-

$\boxed{65CB}$